

Homework 1

Due Friday, January 24

1. The group S_3 discussed in the textbook can be viewed as a group of rotations, in the 3-dimensional space, that are symmetries of an equilateral triangle. Denote by r the $2\pi/3$ rotation about the 3rd-order axis perpendicular to the plane of the triangle and by s the π rotation about one of the in-plane 2nd-order axes.
 - (a) Verify that any of the 6 group elements (including $e = \text{no rotation}$) can be written as either r^k or sr^k , where $k = 0, 1, 2$.
 - (b) The result from (a) can be used to compute the conjugacy classes, i.e., the sets formed by $g^{-1}r^k g$ and $g^{-1}sr^k g$ with g running over the entire group. Find these classes (there should be 3 of them).
 - (c) Verify that assigning value 1 to r and either 1 or -1 to s produces two one-dimensional irreps* of the group.
2. For the same group as in Prob. 1, consider a (reducible) representation in the space V of complex vectors $v = (v_1, v_2, v_3)^T$, where the i -th component ($i = 1, 2, 3$) refers to the i -th vertex of the triangle. The action of the group in this representation is as follows: if a group element g replaces vertex i with vertex j , then the corresponding linear operator $D(g)$ replaces the i -th component of v with the j -th.
 - (a) Construct the 3×3 matrices $D(r)$ and $D(s)$ corresponding to r and s in this representation. Please use the basis formed by $e^{(1)} = (1, 0, 0)^T$, $e^{(2)} = (0, 1, 0)^T$, and $e^{(3)} = (0, 0, 1)^T$.
 - (b) The representation space V contains a one-dimensional invariant subspace W equivalent to one of the irreps found in Prob. 1 part (c). Find the general form of a vector $v \in W$.
 - (c) Consider the orthogonal complement $W^\perp$ to the subspace found in (b) (with the usual definition of inner product for complex vectors). Find an orthonormal basis in $W^\perp$ in which the 2×2 matrix corresponding to s is diagonal and obtain matrices for both s and r in that basis.

*Irreducible representations.